Fast Exact Non-Local Means

Reordering the computation, then studying the quality–runtime trade-off

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17.3×	2.3e–13	+1.00 dB	6.57×
mean exact speed-up K=7, S=11	largest direct-vs-fast pixel difference	full D5 over fast exact NLM at σ=50	fusion faster than full D5 at σ=50

SECTION 1

Assignment route and contribution

I chose **Non-Local Means (NLM)** for this assignment. My main contribution is an **output-equivalent acceleration**. Basic NLM repeatedly sums every K×K patch distance at every search displacement. I instead form one squared-difference image for each displacement and aggregate it with a separable, constant-cost box operator.

After verifying that this faster form gives the same output, I explored a second question: can a richer patch representation improve denoising when extra computation is acceptable? That experiment led to the spatial-PCA D5 expert and a selective fusion gate. I report them separately because they trade speed for quality, while the first method is an exact computational improvement.

SECTION 2

Basic NLM and exact acceleration

For noisy image Y, patch P(p), odd patch size K, odd search size S and noise σ, the bias-corrected distance and NLM estimate are

$$d^2(p,q) = \max(K^{-2}\|P(p)-P(q)\|_2^2 - 2\sigma^2, 0)$$
$$\hat{I}(p) = \Sigma_{q\in S(p)} w(p,q) Y(q) \,/\, \Sigma_{q\in S(p)} w(p,q),$$
$$w = \exp[-d^2/h^2]. \qquad (1)$$

Direct form. For each of N pixels, the basic implementation visits S^2 displacements and sums K^2 samples, giving **$O(NS^2K^2)$** time.

Accelerated form. For displacement t, compute $D_t(u) = [Y(u) - Y(u+t)]^2$, then $d^2(p,p+t) = \max(B_K[D_t](p) - 2\sigma^2, 0)$, where B_K is a separable running box mean. Accumulate the same weights and intensities immediately. This is **$O(NS^2)$** time and O(N) working memory; reflect padding preserves boundary context. In other words, I changed the order of the computation, not the estimator.

EQUIVALENCE CHECK

max |Δ| vs direct NLM	**2.3e–13**
verdict	**OUTPUT–IDENTICAL**

Four 96×96 crops at σ=25, K=7, S=11; the difference sits at double-precision round-off.

SECTION 3

D5: an approximate patch expert

D5 standardizes and spatially windows each K×K patch before PCA:

$$x_p = W^{1/2}(P(p) - \mu), \quad z_p = U^T x_p,$$
$$Q(p) = [z_p, \lambda_y y_p, \lambda_x x_p]. \qquad (2)$$

I use a KD-tree to retrieve a bounded candidate set in Q-space, then rerank those candidates with the exact descriptor distance. Gaussian weights combine their **centre intensities (R0)**. PCA keeps the patch description compact, while the coordinate terms prevent the search from ignoring spatial distance. D5 is therefore an approximate NLM variant, not textbook NLM, and its value must be demonstrated by the experiment.

SECTION 4

Selective fusion and gate

Running D5 everywhere is expensive. To reduce that cost, I combine local residual, gradient, variance, and expert-disagreement cues into a difficulty score g(p). A fixed budget sends only the most difficult pixels to D5:

$$\hat{I}_F(p) = m(p)\hat{I}_{D5}(p) + [1-m(p)]\hat{I}_{cheap}(p),$$
$$m(p) = 1[g(p) \text{ is in the top } \rho \text{ fraction}]. \qquad (3)$$

ρ is 0.10, 0.25 and 0.35 for σ=10,25,50. Expert maps, scoring, PCA fitting, retrieval and reconstruction are all included in the fusion timer.

SCOPE OF THE FUSION CLAIM

Fusion reduces the cost of *full D5*; it does not make fast exact NLM faster. Its gains were confirmed with a new noise seed, but only on the pre-screened four-image set. I therefore treat fusion as an extension, not as the main assignment claim.

SECTION 5

Experimental protocol

- I first screened the 39 available USC-SIPI images at 128 px. After removing the gate/calibration images, I kept images where both D5 and fusion beat fast exact NLM in PSNR and SSIM at all three noise levels: 4.1.02, 4.1.01, 7.1.04, and 7.1.03.
- I then reran these four at their native 256×256 or 512×512 size with a new seed, 20261029. Noise σ∈{10,25,50} is clipped to [0,255]. The gate was trained on a separate four-image set with seed 20260827, so it never saw the final images.
- K=7, S=21 for the native-image comparison. Whole-call wall time excludes file I/O and metric calculation; identical machine and environment for all methods.

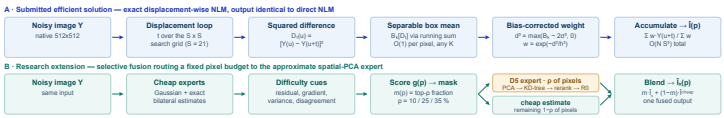


FIGURE 1 · BOTH PIPELINES END TO END. **LANE A**, the submitted solution, changes only the order of evaluation — the estimator is untouched. **LANE B**, the research extension, spends a fixed pixel budget ρ on the approximate expert, so ρ controls real retrieval work rather than merely selecting an output after the fact. The lattice beside each section number marks one displacement in the S×S search grid.

Native-resolution visual and quantitative results

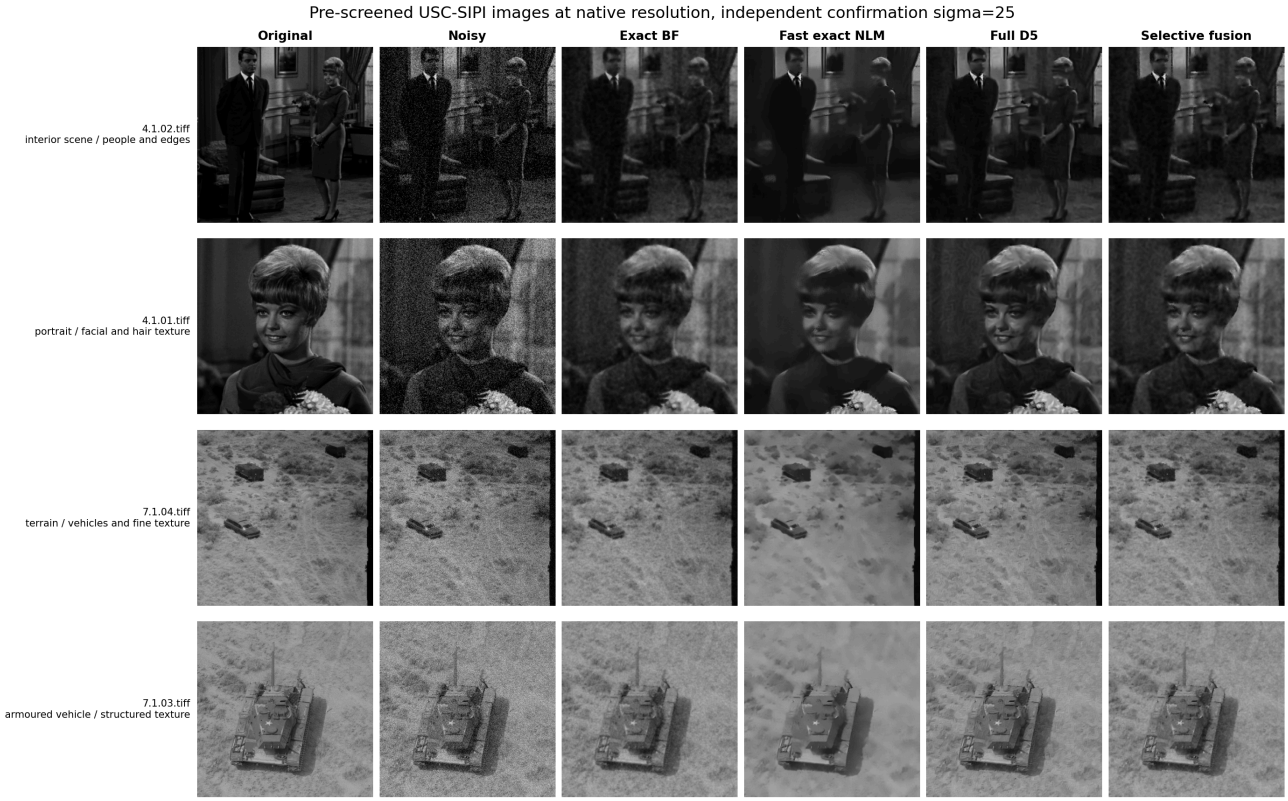


FIGURE 2. Independent native-size confirmation at σ=25. Rows are interior scene, portrait, terrain/vehicles and armoured vehicle. Columns: original, noisy, bilateral, fast exact NLM, full D5, and selective fusion. The set spans faces, edges, fine terrain and structured texture.

TABLE 1. Mean PSNR (dB) / SSIM over four native images. Higher is better; metric-wise best values are bold.

Σ	BILATERAL	FAST EXACT NLM	FULL D5	FUSION
10	33.350 / **0.8573**	33.032 / 0.8466	**33.766** / 0.8573	33.359 / 0.8523
25	29.360 / 0.7285	28.805 / 0.6796	**30.048** / **0.7453**	29.790 / 0.7383
50	25.952 / 0.5957	25.247 / 0.5385	**26.251** / **0.6045**	26.115 / 0.6040

With the new seed, D5 and fusion still exceed fast exact NLM in both metrics at σ=10, 25 and 50. At σ=25, D5 gains 0.936 dB / +0.0654 SSIM on the 512×512 pair and 1.548 dB / +0.0659 SSIM on the 256×256 pair. Because the images were pre-screened, these numbers describe this selected set rather than every possible image.

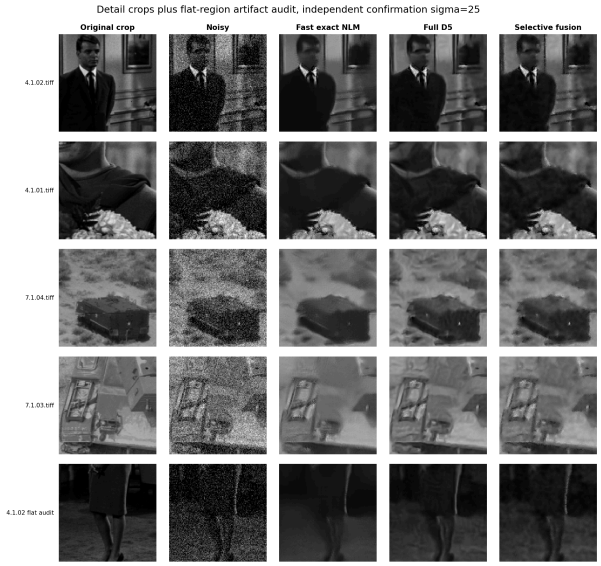


FIGURE 3. Automatically selected high-gradient crops at σ=25 plus a fifth flat-region audit row. The detail crops use maximum clean-image gradient energy; the audit uses minimum gradient energy so structured artifacts cannot hide behind aggregate metrics.

Efficiency evidence, trade-offs, and conclusion

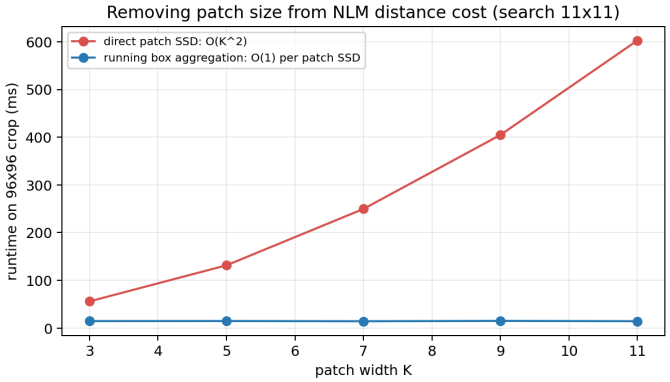


FIGURE 4. Patch-size scaling on a fixed 96×96 crop (S=11). Direct NLM grows approximately with K²; the displacement/box version remains near 15 ms while preserving the same output.

TABLE 2. Exact-equivalence benchmark, K=7, S=11 (ms).

IMAGE	DIRECT	FAST	SPEED-UP	MAX |Δ|
img_002	248.9	14.8	**16.80×**	1.1e−13
img_001	248.6	14.2	**17.52×**	1.4e−13
img_028	250.4	14.7	**17.09×**	2.3e−13
img_027	249.1	14.1	**17.65×**	2.0e−13

Mean speed-up: **17.26×**. At K=11 the measured speed-up reaches **42.15×**, consistent with eliminating the K² factor. This controlled crop test is used because a literal direct native-resolution run would add cost without changing the equivalence claim.

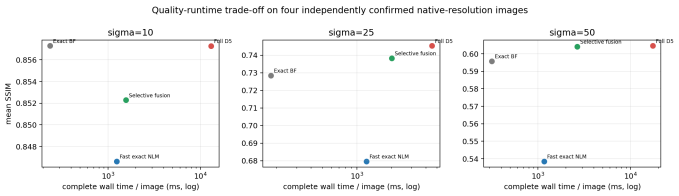


FIGURE 5. Mean quality against whole-call runtime (log scale). D5 and fusion improve selected-set quality, while fast exact NLM remains the substantially cheaper NLM route.

TABLE 3. Mean wall time per image (ms), shown as 256×256 / 512×512.

Σ	BF	FAST NLM	D5	FUSION	D5/FUSION
10	97.6 / 376.4	470.5 / 2007.0	3408.1 / 22633.9	698.8 / 2426.7	4.88× / 9.33×
25	71.8 / 286.2	456.8 / 1972.4	1901.6 / 7135.0	836.7 / 3189.6	2.27× / 2.24×
50	130.5 / 504.9	450.5 / 1886.8	2488.4 / 32345.5	1251.7 / 4049.9	1.99× / 7.99×

All required expert construction and gate computation is included. Runtime values are machine-dependent; relative values within this run are the intended comparison. Each size entry is the mean of two complete native-image calls.

SECTION 6

What succeeded, what did not, and the final claim

The exact acceleration worked. Reusing displacement-wise differences removes the repeated K² patch work without changing the NLM estimate. The measured output is equal to direct NLM up to floating-point precision, with a 17.3× mean speed-up at K=7 and larger gains as K grows.

D5 bought extra quality, but at a clear cost. On these four confirmed images, the spatial-PCA representation and reranked retrieval improve both PSNR and SSIM over fast exact NLM at every noise level. The flat-region crop also shows why I did not rely on the two scalar scores alone: structured texture can still remain visible.

Fusion recovered part of the D5 runtime. It is 8.33×, 2.24× and 6.57× faster than full D5 at the three noise levels. It also improves both metrics over fast exact NLM at all three levels, but remains slower than fast exact NLM. For this set, it is a useful middle ground between the cheaper exact method and full D5. A broader image set and an uncertainty-aware gate would be the next step.

Conclusion. My submitted efficient solution is fast exact NLM. D5 and fusion are supporting experiments: they show how much quality approximate retrieval can add, how much time it costs, and why visual inspection still matters.

References

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